

1 Power Analysis for Longitudinal Clinical Trials with
2 Run-In and Common Close Observations

3 Ronald G. Thomas, Ph.D.*

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5 **Abstract**

6 **Background.** Longitudinal clinical trials often incorporate pre-
7 randomisation (run-in) and post-treatment (common close) observations to
8 improve estimator efficiency. Closed-form power and sample-size formulas
9 for these augmented designs under linear mixed-effects models with random
10 slopes have remained scattered across the literature and partially specified,
11 making applied sample-size calculation cumbersome.

12 **Methods.** We derive closed-form power and sample-size formulas for
13 three nested cases: (i) two run-in observations, (ii) an arbitrary number J_0
14 of run-in observations, and (iii) J_0 run-in plus J_2 common close observations.
15 All derivations proceed via the Woodbury matrix identity applied to the GLS
16 estimator under change-score parameterisation. The formulas are expressed
17 in terms of the residual variance σ^2 , the random-slope variance σ_b^2 , and the
18 observation spacing t , enabling direct computation of required sample sizes.

19 **Results.** Numerical examples using parameters from Alzheimer’s disease
20 neuroimaging trials demonstrate that two run-in observations can reduce
21 required sample sizes by 30-45%, with additional gains from common close
22 observations. The closed-form expressions allow sensitivity analysis with
23 respect to the signal-to-noise ratio σ_b/σ without re-running simulation.

24 **Conclusions.** The Woodbury-based formulas are the simplest available
25 closed-form expressions for the variance of the treatment-effect estimator
26 under random-slopes longitudinal-trial designs with run-in and common close
27 observations. They serve as a foundation for the AR(1) extension (Paper 2),
28 the optimal-allocation problem (Paper 3), and the GLS-versus-ANCOVA-
29 versus-averaged estimator comparison (Paper 4) in this compendium.

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*Wertheim School of Public Health, UCSD; ORCID: 0000-0003-1686-4965

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55 1 Introduction

56 [1] Beyond its traditional roles, the run-in period has a dis-
57 tinct methodological value in rate-of-change trials: estimat-
58 ing individual baseline trajectories to raise power.

- 59 • Conventional run-in uses include identifying compliant partici-
60 pants and washing out prior treatments.
- 61 • When the outcome is a rate of change, run-in observations esti-
62 mate each participant's baseline trajectory.
- 63 • Incorporating those trajectories into the analysis model substan-
64 tially increases statistical power.

65 To be completed by rgt.

66 *The run-in period has served multiple purposes in clinical trial design, from*
67 *identifying compliant participants to washing out prior treatments¹. A dis-*
68 *distinct and methodologically important use has emerged in trials where the*
69 *primary outcome is a rate of change: estimating individual baseline trajec-*
70 *tories that, when incorporated into the analysis model, substantially increase*
71 *statistical power.*

[2] **Frost and colleagues established the mixed-model foundation: run-in observations help most when slope variability is large relative to measurement error.**

- The key insight is that pre-randomization observations inform each participant’s underlying trajectory.
- This reduces the residual variance of the treatment-effect estimator, and the gain grows with the slope-to-noise ratio.
- In Alzheimer’s neuroimaging trials, where that ratio is favorable, they reported sample-size reductions of 30 to 45 percent.

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² established the theoretical foundation for this approach within the linear mixed effects framework. Their key insight was that when between-subject variability in rates of change is large relative to measurement error, pre-randomization observations provide valuable information about each participant’s underlying trajectory, reducing the residual variance of the treatment effect estimator. Working with neuroimaging outcomes in Alzheimer’s disease trials, where the ratio of slope variability to measurement error is particularly favorable, they demonstrated sample size reductions of 30–45%.

[3] **Wang and colleagues extended the framework with two-period models that jointly model run-in and post-randomization data rather than treating run-in as a covariate.**

- Joint modelling of both periods yields additional power gains of up to 15 percent.
- It permits unequal randomization ratios without power loss.
- Run-in observations contribute placebo-equivalent information for all participants.

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³ extended this framework by proposing two-period linear mixed effects models that jointly model run-in and post-randomization data rather than using run-in observations merely as covariates. Their approach yields additional power gains of up to 15% and enables unequal randomization ratios without power loss, since run-in observations effectively contribute placebo-equivalent information for all participants.

[4] **A parallel literature gives closed-form variance and sample-size formulae for the random-coefficient model without a run-in phase, which form the baseline this paper extends.**

- Lu and colleagues derived the cLDA estimator variance, its link to ANCOVA, and the conditions under which the two coincide.
- Hu and colleagues generalized these to random-coefficient models with monotone missing data and variable follow-up, illustrated

with ADNI data.

- These results cover only the $J_0 = 0$ case and serve as the baseline extended here.

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A parallel literature has developed closed-form variance and sample size formulae for the random coefficient regression model without a run-in phase.⁴ derived the variance of the constrained longitudinal data analysis estimator and its relation to analysis of covariance, and⁵ established efficiency conditions under which the two approaches coincide.⁶ generalized these results to random coefficient regression models with monotone missing data and variable follow-up, providing variance, power, and sample size expressions illustrated with data from the Alzheimer’s Disease Neuroimaging Initiative. These formulae apply to designs with $J_0 = 0$ run-in observations and an arbitrary post-randomization visit schedule, and serve as the baseline case that we extend here.

[5] **The CRAN longpower package is the reference implementation of the baseline formulae, and our general result reduces to its routines at the appropriate boundaries.**

- longpower provides edland, diggle, and liu.liang routines for slope-comparison trials.
- Our formula reduces to the Ard-Edland slope variance at $J_0 = J_2 = 0$, and to the Diggle compound-symmetry form when also $\sigma_b^2 = 0$.
- The generalization thus embeds the existing longpower formulae as the baseline against which run-in and common-close gains are measured.

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The CRAN reference implementation of the $J_0 = 0$ formulae is the longpower package⁷, which provides the routines `edland.linear.power()`, `diggle.linear.power()`, and `liu.liang.linear.power()` for slope-comparison trials with random intercepts and slopes^{8–10}. We show in Section 3.3 that the closed-form result derived below reduces to the Ard-Edland slope-variance formula implemented in `edland.linear.power()` at the boundary $J_0 = J_2 = 0$, and to the Diggle compound-symmetry formula when additionally $\sigma_b^2 = 0$. The generalization developed here therefore embeds the existing longpower formulae as the baseline case against which the efficiency gains from run-in and common close observations are measured.

[6] **No prior closed-form power formula handles an arbitrary number of run-in observations together with a separate common-close phase, the gap this paper fills.**

- The common-close phase observes all participants off-treatment after the randomized period.
- It supplies extra information about individual trajectories at no cost to the treatment comparison.

- Such post-treatment periods appear in delayed-start and randomized-withdrawal designs, but their GLS efficiency contribution has not been formally characterized.

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Despite these advances, no general closed-form power formula has been derived that simultaneously accounts for an arbitrary number of run-in observations and a separate set of common close observations collected after treatment cessation. The common close phase, in which all participants are observed off-treatment following the randomized period, provides additional information about individual trajectories at no cost to the treatment comparison. Post-treatment observation periods appear in delayed-start and randomized withdrawal designs aimed at distinguishing symptomatic from disease-modifying effects¹¹, but the contribution of the common close to statistical efficiency within the GLS framework has not been formally characterized.

[7] **The paper presents three nested results, all derived from the GLS estimator via the Woodbury identity.**

- A closed-form variance for exactly two run-in observations (Section 3.2), then its generalization to arbitrary J_0 (Section 3.3).
- A further extension adding J_2 common-close observations (Section 3.4).
- All expressions are in terms of σ^2 , σ_b^2 , and spacing t , with numerical examples from Alzheimer’s neuroimaging studies (Section 5).

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We present three results. First, we derive a simple closed-form variance formula for the treatment effect estimator when exactly two run-in observations precede randomization (Section 3.2). Second, we generalize this formula to an arbitrary number J_0 of run-in observations (Section 3.3). Third, we further extend to include J_2 common close observations collected after treatment cessation (Section 3.4). All derivations proceed from the generalized least squares estimator under the Woodbury matrix identity, yielding expressions in terms of the residual variance σ^2 , the random slope variance σ_b^2 , and the observation spacing t . Section 5 provides numerical examples using parameters estimated from Alzheimer’s disease neuroimaging studies.

2 Model

2.1 General formulation

[8] **The model accommodates a two-arm longitudinal trial whose observation schedule spans three distinct phases.**

- N participants are observed at equally spaced time points.
- The phases are run-in (pre-randomization), treatment (post-randomization), and an optional common close (post-treatment).
- Y_{ij} denotes the outcome for participant i at time point j .

199 To be completed by rgt.

200 Consider a two-arm longitudinal clinical trial in which N participants are
 201 observed at equally spaced time points across three phases: a run-in period
 202 (pre-randomization), a treatment period (post-randomization), and an op-
 203 tional common close period (post-treatment). Let Y_{ij} denote the outcome for
 204 participant i at time point j . The observation schedule is:

- 205 • **Run-in:** $j = -J_0, -J_0 + 1, \dots, -1$ (all participants untreated)
- 206 • **Randomization visit:** $j = 0$ (baseline)
- 207 • **Treatment period:** $j = 1, 2, \dots, J_1$ (participants randomized to treatment
 208 or placebo)
- 209 • **Common close:** $j = J_1 + 1, J_1 + 2, \dots, J_1 + J_2$ (all participants off-treatment)

210 We model the outcome as

$$Y_{ij} = \alpha + a_i + (\delta(1 - h_j) + \beta h_j + \gamma h_j g_i + b_i)t_j + u_{ij} + \varepsilon_{ij} \quad (1)$$

211 [9] **The outcome model combines a random intercept and ran-**
 212 **dom slope with a three-component within-subject variance**
 213 **structure following Frost and colleagues.**

- 214 • a_i and b_i are the random intercept and slope, while u_{ij} and ε_{ij}
 215 are the between-visit and between-scan residuals.
- 216 • Both residual terms are independent across visits, so under a
 217 single scan per visit they enter every formula only through their
 218 sum.
- 219 • We therefore define the total within-visit residual $w_{ij} = u_{ij} + \varepsilon_{ij}$
 220 with variance σ^2 .

221 To be completed by rgt.

222 where α is the overall intercept, $a_i \sim N(0, \sigma_a^2)$ is a random intercept, $b_i \sim$
 223 $N(0, \sigma_b^2)$ is a random slope, $u_{ij} \sim N(0, \sigma_u^2)$ are within-subject between-visit
 224 random effects independent across visits, and $\varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2)$ are indepen-
 225 dent between-scan measurement errors. This three-component within-subject
 226 variance structure follows² (their Eq. 17). Because u_{ij} and ε_{ij} are both inde-
 227 pendent across visits, under a single scan per visit they enter every formula
 228 below only through their sum. We therefore write $w_{ij} = u_{ij} + \varepsilon_{ij}$ for the total
 229 within-visit residual, with

$$\sigma^2 \equiv \text{Var}(w_{ij}) = \sigma_u^2 + \sigma_\varepsilon^2. \quad (2)$$

230 [10] **The two within-visit components are identifiable only**
 231 **with replicate scans, so single-scan designs depend solely on**
 232 **their sum.**

- 233 • Separating σ_u^2 from σ_ε^2 requires a visit carrying replicate scans.
- 234 • For the single-scan designs treated here, only the sum σ^2 enters
 235 the treatment-effect variance.
- 236 • This reduction simplifies all subsequent variance expressions.

237 To be completed by rgt.

238 *The two within-visit components are separately identifiable only when a visit*
 239 *carries replicate scans; for the single-scan designs treated here only their sum*
 240 *σ^2 enters the variance of the treatment-effect estimator. The phase indicator*
 241 *is*

$$h_j = \begin{cases} 0 & \text{if } j \leq 0 \quad (\text{run-in}) \\ 1 & \text{if } j > 0 \quad (\text{post-randomization}) \end{cases}$$

242 and the treatment group indicator is

$$g_i = \begin{cases} 1 & i = 1, \dots, m \quad (\text{treatment}) \\ 0 & i = m + 1, \dots, N \quad (\text{placebo}) \end{cases}$$

243 [11] **Three fixed-effect slope parameters describe the run-in,**
 244 **placebo post-randomization, and treatment trajectories un-**
 245 **der equally spaced observations.**

- 246 • δ is the run-in rate of change common to all participants.
- 247 • β is the placebo post-randomization rate of change, and γ is the
- 248 treatment effect on that rate.
- 249 • Observations are equally spaced with interval t , so $t_j = jt$.

250 To be completed by rgt.

251 *The fixed effects are: δ , the rate of change during the run-in period (common*
 252 *to all participants); β , the rate of change during the post-randomization period*
 253 *for the placebo group; and γ , the treatment effect on the rate of change. We*
 254 *assume equally spaced observations with interval t , so $t_j = jt$.*

255 [12] **During the common close both arms revert to a shared**
 256 **off-treatment slope, modelled by deactivating the treatment**
 257 **indicator in that phase.**

- 258 • For $j > J_1$ all participants are off-treatment with slope $\beta + b_i$.
- 259 • This is implemented by setting $g_i = 0$ for every participant during
- 260 the common close.
- 261 • Equivalently, the treatment indicator is replaced with $g_i \cdot I(1 \leq$
 262 $j \leq J_1)$ in the model.

263 To be completed by rgt.

264 *For the common close phase ($j > J_1$), both treatment and placebo groups*
 265 *revert to a common off-treatment slope. We model this by setting $g_i = 0$ for*
 266 *all participants during the common close, so that the slope for both groups is*
 267 *$\beta + b_i$ in this phase. This is equivalent to replacing the treatment indicator*
 268 *with $g_i \cdot I(1 \leq j \leq J_1)$ in model (1).*

2.2 Change-score parameterization

Following², we work with change scores relative to the randomization visit:

$$c_{ij} = Y_{ij} - Y_{i0} \quad (3)$$

[13] **Differencing relative to baseline removes the intercept and aligns the analysis with constrained longitudinal data analysis under compound symmetry.**

- The change score eliminates α and the random intercept a_i , leaving only two variance components.
- It is algebraically equivalent to cLDA, so the derived formulas apply to both analyses of run-in designs.
- By dropping σ_a^2 it avoids the anticonservative sample sizes that arise from ignoring intercept-slope correlation.

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This eliminates the intercept α and the random intercept a_i . We note that the change-score approach is algebraically equivalent to constrained longitudinal data analysis (cLDA) under compound symmetry, where the baseline is included in the response vector with a common-mean constraint across treatment groups^{12,13}. The variance formulas derived below therefore apply to both change-score and cLDA analyses of run-in designs. Because the change-score eliminates a_i , the resulting formulas depend on only two variance components (σ^2, σ_b^2) rather than the three ($\sigma^2, \sigma_a^2, \sigma_b^2$) required by the raw-outcome model.¹⁴ show that ignoring the intercept-slope correlation in the three-component model can produce anticonservative sample sizes; the change-score approach avoids this issue by construction, at the cost of discarding information about σ_a^2 .

The change-score formulation yields

$$c_{ij} = (\delta(1 - h_j) + \beta h_j + \gamma h_j g_i + b_i) t_j + (w_{ij} - w_{i0}) \quad (4)$$

for $j \neq 0$. The change-score vector for participant i has dimension $J = J_0 + J_1 + J_2$, encompassing all three phases.

2.3 Covariance structure

The marginal covariance of the change-score vector is

$$\Sigma = R + ZGZ' \quad (5)$$

where $G = \sigma_b^2$ (scalar, since we have a single random slope), $Z = t \cdot \mathbf{1}_J$ is the $J \times 1$ random effects design vector, and R is the $J \times J$ residual covariance of the differenced within-visit residuals ($w_{ij} - w_{i0}$).

301 The residual covariance has the structure

$$R_{jl} = \text{Cov}(w_{ij} - w_{i0}, w_{il} - w_{i0}) = \begin{cases} 2\sigma^2 & \text{if } j = l \\ \sigma^2 & \text{if } j \neq l \end{cases} \quad (6)$$

302 [14] The shared baseline residual induces compound-
 303 symmetry correlation among change scores, so the residual
 304 covariance depends only on the summed within-visit vari-
 305 ance.

- 306 • The within-visit residuals are independent across visits with com-
 307 mon variance σ^2 .
- 308 • Differencing against the shared baseline w_{i0} produces positive
 309 correlation, giving $R = \sigma^2(\mathbf{I}_J + \mathbf{1}_J\mathbf{1}_J')$.
- 310 • Both within-visit components enter R identically, so only their
 311 sum is identified from single-scan designs.

312 To be completed by rgt.

313 since $w_{ij} = u_{ij} + \varepsilon_{ij}$ are independent across visits with common variance
 314 $\sigma^2 = \sigma_u^2 + \sigma_\varepsilon^2$, and the shared baseline residual w_{i0} induces positive correla-
 315 tion among all change scores. Thus $R = \sigma^2(\mathbf{I}_J + \mathbf{1}_J\mathbf{1}_J')$. Both within-visit
 316 components enter R identically because u_{i0} and ε_{i0} are differenced out of ev-
 317 ery change score in the same way; this is why only their sum σ^2 is identified
 318 from single-scan designs.

319 3 Variance of the Treatment Effect Estimator

320 3.1 GLS estimation and the Woodbury identity

321 The generalized least squares estimator of the fixed effects is

$$\hat{\beta} = (X'\Sigma^{-1}X)^{-1}X'\Sigma^{-1}\mathbf{c} \quad (7)$$

322 [15] The estimand of interest is the treatment-effect variance,
 323 a single diagonal element of the inverted GLS information
 324 matrix.

- 325 • The GLS estimator has covariance $(X'\Sigma^{-1}X)^{-1}$.
- 326 • X is the fixed-effects design matrix and $\mathbf{c}$ is the stacked change-
 327 score vector.
- 328 • The target $\text{Var}(\hat{\gamma})$ is the diagonal entry corresponding to γ .

329 To be completed by rgt.

330 with $\text{Var}(\hat{\beta}) = (X'\Sigma^{-1}X)^{-1}$, where X is the fixed effects design matrix and
 331 $\mathbf{c}$ is the stacked change-score vector. Our target is $\text{Var}(\hat{\gamma})$, the variance of
 332 the treatment effect estimator, which is the diagonal element of $(X'\Sigma^{-1}X)^{-1}$
 333 corresponding to γ .

334 The key computational tool is the Woodbury matrix identity. Since $\Sigma = R + ZGZ'$

335 with scalar $G = \sigma_b^2$:

$$\Sigma^{-1} = R^{-1} - R^{-1}Z(G^{-1} + Z'R^{-1}Z)^{-1}Z'R^{-1} \quad (8)$$

336 This avoids direct inversion of the $J \times J$ matrix Σ and enables closed-form results
337 because R^{-1} has a known simple structure.

338 From $R = \sigma^2(\mathbf{I}_J + \mathbf{1}_J\mathbf{1}_J')$, the Sherman-Morrison formula gives

$$R^{-1} = \frac{1}{\sigma^2} \left(\mathbf{I}_J - \frac{1}{J+1} \mathbf{1}_J \mathbf{1}_J' \right) \quad (9)$$

339 With $Z = t \cdot \mathbf{1}_J$, we obtain

$$Z'R^{-1}Z = \frac{t^2}{\sigma^2} \left(J - \frac{J^2}{J+1} \right) = \frac{J t^2}{\sigma^2(J+1)} \quad (10)$$

340 and

$$(G^{-1} + Z'R^{-1}Z)^{-1} = \frac{\sigma^2 \sigma_b^2 (J+1)}{\sigma^2(J+1) + J t^2 \sigma_b^2} \quad (11)$$

341 Let $a = \sigma^2$ and $b = \sigma_b^2$ for conciseness. Define

$$\eta_J = \frac{a(J+1) + J t^2 b}{a(J+1)} = 1 + \frac{J t^2 b}{a(J+1)} \quad (12)$$

342 so that $(G^{-1} + Z'R^{-1}Z)^{-1} = b/\eta_J$.

343 3.1.1 Per-participant information, compact form

344 Let X_i , Z_i , and R_i denote the per-participant fixed effects design, random effects
345 design, and residual covariance, and define the scalar

$$\mathcal{J}_i = Z_i' R_i^{-1} Z_i + G^{-1} \quad (13)$$

346 By (8), the per-participant contribution to the GLS information matrix is

$$X_i' \Sigma_i^{-1} X_i = X_i' R_i^{-1} X_i - \frac{(X_i' R_i^{-1} Z_i)(Z_i' R_i^{-1} X_i)}{\mathcal{J}_i} \quad (14)$$

347 [16] **The additive per-participant information form has two**
348 **properties exploited later: a scalar rank-one correction and**
349 **accommodation of heterogeneous designs.**

- 350 • The total information is the sum of per-participant contributions,
351 with covariance given by its inverse.
- 352 • Under a single random slope, $\mathcal{J}_i$ is a scalar, making the correction
353 a rank-one adjustment of the OLS-style information.
- 354 • Because contributions add, varying J_0 , J_2 , or dropout patterns
355 are handled by summing over strata.

356 To be completed by rgt.

357 and the total information is $X'\Sigma^{-1}X = \sum_i X_i'\Sigma_i^{-1}X_i$, with $\text{Var}(\hat{\beta}) =$
 358 $(X'\Sigma^{-1}X)^{-1}$. Two features of this form are used later: (i) under a single
 359 random slope, $\mathcal{J}_i$ is a scalar so (14) is a rank-one correction of the OLS-style
 360 information $X_i'R_i^{-1}X_i$; (ii) because per-participant contributions add, het-
 361 erogeneous designs (varying J_0 , J_2 , or dropout patterns) are accommodated
 362 by summing (14) over the relevant strata.

363 3.2 Case 1: Standard design, no run-in ($J_0 = 0$, $J_1 = 2$, $J_2 = 0$)

364 [17] **The no-run-in case reproduces established results and**
 365 **serves as a verification baseline for the framework.**

- 366 • It reproduces Frost and colleagues, Section 3.1.1.
- 367 • It is the $J_0 = 0$ baseline of the random-coefficient power formulae
 368 of Lu and Hu and colleagues.
- 369 • It thereby provides a verification checkpoint for the present
 370 derivation.

371 To be completed by rgt.

372 *This case reproduces the result in², Section 3.1.1, and is the $J_0 = 0$ baseline*
 373 *of the random coefficient regression power formulae of⁴ and⁶. It serves as*
 374 *verification for our framework.*

375 [18] **Without a run-in phase the design reduces to two**
 376 **post-randomization change scores and a two-parameter**
 377 **fixed-effects model.**

- 378 • There are $J = 2$ change scores at post-randomization times t and
 379 $2t$.
- 380 • The absence of run-in removes the δ parameter from the model.
- 381 • The remaining fixed effects are the placebo slope β and the treat-
 382 ment effect γ .

383 To be completed by rgt.

384 *With $J_0 = 0$ and $J_1 = 2$, there are $J = 2$ change scores ($c_{i,1}$, $c_{i,2}$) at post-*
 385 *randomization times (t , $2t$). Since there is no run-in, the model has two fixed*
 386 *effects: the placebo slope β and the treatment effect γ .*

387 The design matrices are:

$$X_i^{(\text{trt})} = t \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix}, \quad X_i^{(\text{plc})} = t \begin{pmatrix} 1 & 0 \\ 2 & 0 \end{pmatrix}$$

388 The residual covariance is $R = a \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ with $R^{-1} = \frac{1}{3a} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$, and the
 389 random effects design is $Z = t(1, 2)'$.

390 Applying the Woodbury identity with $J = 2$:

$$Z'R^{-1}Z = \frac{t^2}{3a}(1, 2) \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \frac{t^2}{3a}(1, 2)(0, 3)' = \frac{6t^2}{3a} = \frac{2t^2}{a} \quad (15)$$

391

$$(G^{-1} + Z'R^{-1}Z)^{-1} = \frac{ab}{a + 2bt^2} \quad (16)$$

392 After summing over one treatment and one placebo participant, the information
393 matrix is

$$X'\Sigma^{-1}X = \frac{2t^2}{a + 2bt^2} \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \quad (17)$$

394 and its inverse is

$$\text{Var}(\hat{\beta}) = \frac{a + 2bt^2}{2t^2} \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \quad (18)$$

395 The variance of the treatment effect estimator (per participant pair) is the (2, 2)
396 element:

$$\boxed{\text{Var}(\hat{\gamma})|_{J_0=0, J_1=2} = \frac{a + 2bt^2}{t^2} = \frac{\sigma^2 + 2\sigma_b^2 t^2}{t^2}} \quad (19)$$

397 **[19] The recovered variance matches Frost and supplies the**
398 **per-group baseline for all run-in comparisons.**

- 399 • The expression agrees with Frost and colleagues, Section 3.1.1.
- 400 • For m participants per group the variance scales as $(\sigma^2 +$
401 $2\sigma_b^2 t^2)/(mt^2)$.
- 402 • This quantity is the reference against which run-in designs are
403 evaluated.

404 *To be completed by rgt.*

405 *This agrees with Frost et al. (2008), Section 3.1.1. For m participants per*
406 *group: $\text{Var}(\hat{\gamma}) = (\sigma^2 + 2\sigma_b^2 t^2)/(mt^2)$. This serves as the baseline against*
407 *which the run-in designs are compared.*

408 3.3 Case 2: Two run-in observations ($J_0 = 2, J_1 = 1, J_2 = 0$)

409 **[20] Two run-in observations yield three change scores and a**
410 **design matrix that distinguishes the run-in slope from the**
411 **post-randomization slopes.**

- 412 • There are $J = 3$ change scores spanning the pre- and post-
413 randomization times.
- 414 • The fixed-effects model now includes the run-in slope δ alongside
415 β and γ .
- 416 • The design matrix separates the run-in contribution from the
417 post-randomization contribution.

418 *To be completed by rgt.*

419 We now have $J = 3$ change scores $(c_{i,-2}, c_{i,-1}, c_{i,1})$ at times $(-2t, -t, 0, t)$.
 420 The fixed effects design matrix separates the run-in slope δ from the post-
 421 randomization slopes:

$$X_i^{(\text{trt})} = \begin{pmatrix} -2t & 0 & 0 \\ -t & 0 & 0 \\ 0 & t & t \end{pmatrix}, \quad X_i^{(\text{plc})} = \begin{pmatrix} -2t & 0 & 0 \\ -t & 0 & 0 \\ 0 & t & 0 \end{pmatrix}$$

422 [21] Because the random slope acts uniformly across visits,
 423 the random-effects design vector collapses to the scaled time
 424 distances from baseline.

- 425 • The random slope b_i applies to all time points alike.
- 426 • The random-effects design therefore reduces to $Z = t(-2, -1, 1)'$.
- 427 • The run-in change scores are differences of the pre-randomization
- 428 outcomes from baseline.

429 To be completed by rgt.

430 The random effects design is $Z = t(1, 1, 1)' \cdot (-2, -1, 1)' \cdot b_i$. However, since
 431 the random slope applies uniformly to all time points, $Z = t(-2, -1, 1)'$ for
 432 the random effect b_i . We note that the change scores at times $-2t$ and $-t$
 433 relative to baseline at 0 yield $c_{i,-2} = Y_{i,-2} - Y_{i,0}$ and $c_{i,-1} = Y_{i,-1} - Y_{i,0}$.

434 More carefully, under model (1):

$$\begin{aligned} c_{i,-2} &= -2(\delta + b_i)t + (w_{i,-2} - w_{i,0}) \\ c_{i,-1} &= -(\delta + b_i)t + (w_{i,-1} - w_{i,0}) \\ c_{i,1} &= (\beta + \gamma g_i + b_i)t + (w_{i,1} - w_{i,0}) \end{aligned} \quad (20)$$

435 Thus the design matrices for the fixed effects $(\delta, \beta, \gamma)'$ are:

$$X_i^{(\text{trt})} = t \begin{pmatrix} -2 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \quad X_i^{(\text{plc})} = t \begin{pmatrix} -2 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

436 and the random effects design is $Z_i = t(-2, -1, 1)'$.

437 The residual covariance is

$$R = a \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix} = a(\mathbf{I}_3 + \mathbf{1}_3 \mathbf{1}_3') \quad (21)$$

438 with inverse

$$R^{-1} = \frac{1}{a} \left(\mathbf{I}_3 - \frac{1}{4} \mathbf{1}_3 \mathbf{1}_3' \right) = \frac{1}{4a} \begin{pmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{pmatrix} \quad (22)$$

439 Now we compute the Woodbury components. With $Z = t(-2, -1, 1)'$:

$$Z'R^{-1}Z = \frac{t^2}{4a}(3 \cdot 4 + 3 \cdot 1 + 3 \cdot 1 - 2(-1)(-2)(-1) - 2(-1)(-2)(1) - 2(-1)(-1)(1)) \quad (23)$$

440 Computing directly: $Z'R^{-1}Z = \frac{t^2}{4a}(-2, -1, 1) \begin{pmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{pmatrix} (-2, -1, 1)'$

$$R^{-1}Z = \frac{t}{4a} \begin{pmatrix} 3(-2) + (-1)(-1) + (-1)(1) \\ (-1)(-2) + 3(-1) + (-1)(1) \\ (-1)(-2) + (-1)(-1) + 3(1) \end{pmatrix} = \frac{t}{4a} \begin{pmatrix} -6 \\ -2 \\ 6 \end{pmatrix}$$

441 so $Z'R^{-1}Z = \frac{t^2}{4a}[(-2)(-6) + (-1)(-2) + (1)(6)] = \frac{t^2}{4a}(12 + 2 + 6) = \frac{20t^2}{4a} = \frac{5t^2}{a}$.

442 Alternatively, we note $Z = t \cdot \mathbf{d}$ where $\mathbf{d} = (-2, -1, 1)'$, so $Z'R^{-1}Z = \frac{t^2}{a}(\mathbf{d}'\mathbf{d} -$
443 $\frac{1}{4}(\mathbf{1}'\mathbf{d})^2) = \frac{t^2}{a}(6 - \frac{4}{4}) = \frac{5t^2}{a}$.

444 Thus

$$H \equiv (G^{-1} + Z'R^{-1}Z)^{-1} = \frac{ab}{a + 5bt^2} \quad (24)$$

445 For the information matrix, we compute $X'R^{-1}X$ summed over treatment and
446 placebo groups. For a single treatment participant:

$$X_i^{(\text{trt})'}R^{-1}X_i^{(\text{trt})} = \frac{t^2}{4a} \begin{pmatrix} -2 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{pmatrix} \begin{pmatrix} -2 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

447 Computing the product:

$$X_i^{(\text{trt})'}R^{-1}X_i^{(\text{trt})} = \frac{t^2}{4a} \begin{pmatrix} 11 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{pmatrix} \quad (25)$$

448 For placebo (whose γ column is zero):

$$X_i^{(\text{plc})'}R^{-1}X_i^{(\text{plc})} = \frac{t^2}{4a} \begin{pmatrix} 11 & 3 & 0 \\ 3 & 3 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (26)$$

449 Summing (one participant per group for the derivation, then multiplying by m):

$$X'R^{-1}X = \frac{t^2}{4a} \begin{pmatrix} 22 & 6 & 3 \\ 6 & 6 & 3 \\ 3 & 3 & 3 \end{pmatrix} \quad (27)$$

450 For the Woodbury correction, we need $X'R^{-1}Z$. For treatment:

$$X_i^{(\text{trt})'} R^{-1} Z = \frac{t^2}{4a} \begin{pmatrix} -2 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -6 \\ -2 \\ 6 \end{pmatrix} = \frac{t^2}{4a} \begin{pmatrix} 14 \\ 6 \\ 6 \end{pmatrix}$$

451 For placebo:

$$X_i^{(\text{plc})'} R^{-1} Z = \frac{t^2}{4a} \begin{pmatrix} -2 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} -6 \\ -2 \\ 6 \end{pmatrix} = \frac{t^2}{4a} \begin{pmatrix} 14 \\ 6 \\ 0 \end{pmatrix}$$

452 The Woodbury correction requires the per-subject outer products (not the outer
453 product of the sum):

$$\begin{aligned} W &= \sum_g X_i^{(g)'} R^{-1} Z \cdot H \cdot Z' R^{-1} X_i^{(g)} \\ &= \frac{bt^4}{16a(a + 5bt^2)} \left(\begin{pmatrix} 14 \\ 6 \\ 6 \end{pmatrix} (14, 6, 6) + \begin{pmatrix} 14 \\ 6 \\ 0 \end{pmatrix} (14, 6, 0) \right) \\ &= \frac{bt^4}{16a(a + 5bt^2)} \begin{pmatrix} 392 & 168 & 84 \\ 168 & 72 & 36 \\ 84 & 36 & 36 \end{pmatrix} \end{aligned} \quad (28)$$

454 Therefore

$$X'\Sigma^{-1}X = X'R^{-1}X - W \quad (29)$$

455 **[22] The treatment-effect variance is the relevant diagonal el-**
456 **ement of the inverted information matrix, obtained numeri-**
457 **cally and checked against the closed form.**

- 458 • $\text{Var}(\hat{\gamma})$ is the $(3, 3)$ element of $(X'\Sigma^{-1}X)^{-1}$.
- 459 • It is evaluated numerically in the accompanying R code.
- 460 • The numerical value is verified against the derived closed-form
- 461 result.

462 *To be completed by rgt.*

463 *To find $\text{Var}(\hat{\gamma})$, we need the $(3, 3)$ element of $(X'\Sigma^{-1}X)^{-1}$. We compute this*
464 *numerically in the R code below and verify against the closed-form result.*

465 **[23] Completing the cofactor and determinant algebra yields**
466 **the closed-form treatment-effect variance for the two-run-in**
467 **design.**

- 468 • Define the auxiliary quantity $D = \sigma^2 + 5\sigma_b^2 t^2$.
- 469 • The $(3, 3)$ cofactor simplifies to $6t^4/(aD)$.
- 470 • Dividing the cofactor by the determinant gives the boxed variance

below.

To be completed by rgt.

Carrying through the algebra, define $D = a + 5bt^2 = \sigma^2 + 5\sigma_b^2 t^2$ and write $X'\Sigma^{-1}X = \frac{t^2}{16aD} A$. The matrix A has $(3,3)$ minor $1536aD$ and determinant $9216aD(a + 2bt^2)$. Hence the $(3,3)$ cofactor of $X'\Sigma^{-1}X$ is $1536aD \cdot t^4/(16aD)^2 = 6t^4/(aD)$, and its determinant is $9216aD(a + 2bt^2) \cdot t^6/(16aD)^3$. The ratio gives:

$$\text{Var}(\hat{\gamma})|_{J_0=2, J_1=1} = \frac{8\sigma^2(\sigma^2 + 5\sigma_b^2 t^2)}{3t^2(\sigma^2 + 2\sigma_b^2 t^2)} \quad (30)$$

[24] The run-in variance differs structurally from the Frost baseline by involving the residual variance in both numerator and denominator.

- In the Frost formula σ^2 appears only additively in the numerator.
- Here σ^2 enters both the numerator and the denominator.
- This reflects the extra information extracted from pre-randomization observations.

To be completed by rgt.

Unlike the Frost formula (19), the run-in variance depends on σ^2 both in the numerator and denominator, reflecting the additional information extracted from pre-randomization observations.

3.4 General formula: J_0 run-in observations ($J_2 = 0$)

[25] The derivation now generalizes to an arbitrary number of run-in observations with no common close.

- The design has arbitrary J_0 run-in and J_1 post-randomization observations.
- The common close is absent, so $J_2 = 0$.
- The total number of change scores is $J = J_0 + J_1$.

To be completed by rgt.

We now derive the variance for arbitrary J_0 run-in observations with J_1 post-randomization observations and no common close ($J_2 = 0$). The total number of change scores is $J = J_0 + J_1$.

The change scores are

$$\begin{aligned} c_{i,-s} &= -s(\delta + b_i)t + (w_{i,-s} - w_{i0}) & s &= 1, \dots, J_0 \\ c_{i,l} &= l(\beta + \gamma g_i + b_i)t + (w_{i,l} - w_{i0}) & l &= 1, \dots, J_1 \end{aligned} \quad (31)$$

[26] The design matrix is built from run-in and post-randomization time vectors mapping to the three fixed-effect columns.

- The columns correspond to the parameters δ , β , and γ .
- $\mathbf{d}_{J_0}$ collects the negative run-in times scaled by t .
- $\mathbf{d}_{J_1}$ collects the positive post-randomization times scaled by t .

To be completed by rgt.

The fixed effects design matrix has columns for (δ, β, γ) . Let $\mathbf{d}_{J_0} = (-J_0, -J_0 + 1, \dots, -1)' \cdot t$ be the run-in time vector and $\mathbf{d}_{J_1} = (1, 2, \dots, J_1)' \cdot t$ be the post-randomization time vector. Then:

$$X_i^{(\text{trt})} = \begin{pmatrix} \mathbf{d}_{J_0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{d}_{J_1} & \mathbf{d}_{J_1} \end{pmatrix}, \quad X_i^{(\text{plc})} = \begin{pmatrix} \mathbf{d}_{J_0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{d}_{J_1} & \mathbf{0} \end{pmatrix}$$

The random effects design vector is $Z_i = (\mathbf{d}_{J_0}', \mathbf{d}_{J_1}')'$, i.e., the full vector of time distances from baseline.

The residual covariance remains $R = a(\mathbf{I}_J + \mathbf{1}_J \mathbf{1}_J')$ with inverse given by (9).

3.4.1 Computing $X' \Sigma^{-1} X$

[27] The Woodbury computation is organized around the sum-of-squares of the run-in and post-randomization time indices.

- The derivation proceeds via the Woodbury identity.
- S_{J_0} is the sum of squared run-in indices in closed form.
- S_{J_1} is the corresponding sum over post-randomization indices.

To be completed by rgt.

We proceed through the Woodbury identity. Define $S_{J_0} = \sum_{s=1}^{J_0} s^2 = J_0(J_0 + 1)(2J_0 + 1)/6$ and $S_{J_1} = \sum_{l=1}^{J_1} l^2 = J_1(J_1 + 1)(2J_1 + 1)/6$.

The key intermediate quantities are:

$Z' R^{-1} Z$: With $Z = t \cdot \mathbf{d}$ where $\mathbf{d} = (-J_0, \dots, -1, 1, \dots, J_1)'$:

$$Z' R^{-1} Z = \frac{t^2}{a} \left[(S_{J_0} + S_{J_1}) - \frac{(\sum d_j)^2}{J + 1} \right] \quad (32)$$

where $\sum d_j = -J_0(J_0 + 1)/2 + J_1(J_1 + 1)/2$.

$(G^{-1} + Z' R^{-1} Z)^{-1}$: From (32):

$$H = \frac{ab}{a + t^2 b \cdot Z' R^{-1} Z \cdot a / t^2}$$

$X' R^{-1} X$: This 3×3 matrix has elements computed from the block structure of X . Letting $D_{J_0} = \sum_{s=1}^{J_0} s = J_0(J_0 + 1)/2$ and $D_{J_1} = \sum_{l=1}^{J_1} l = J_1(J_1 + 1)/2$, and using $R^{-1} = \frac{1}{a}(\mathbf{I}_J - \frac{1}{J+1} \mathbf{1} \mathbf{1}')$, the information matrix summed over one treatment and one placebo participant is:

$$X'R^{-1}X = \frac{t^2}{a} \begin{pmatrix} 2\left(S_{J_0} - \frac{D_{J_0}^2}{J+1}\right) & \frac{2D_{J_0}D_{J_1}}{J+1} & \frac{D_{J_0}D_{J_1}}{J+1} \\ \frac{2D_{J_0}D_{J_1}}{J+1} & 2\left(S_{J_1} - \frac{D_{J_1}^2}{J+1}\right) & S_{J_1} - \frac{D_{J_1}^2}{J+1} \\ \frac{D_{J_0}D_{J_1}}{J+1} & S_{J_1} - \frac{D_{J_1}^2}{J+1} & S_{J_1} - \frac{D_{J_1}^2}{J+1} \end{pmatrix} \quad (33)$$

[28] **The structure of the information matrix reflects which parameters both groups inform and the opposing signs of the two phase time-sums.**

- Entries informed by both groups carry a factor of 2.
- The γ -related entries lack this factor because the placebo γ column is zero.
- The positive (δ, β) cross-term arises from the opposite signs of the run-in and post-randomization time sums.

To be completed by rgt.

The factor of 2 appears on entries involving both groups, while γ -related entries lack this factor since the γ column is zero for the placebo group. The positive sign on the (δ, β) cross-term arises because the run-in and post-randomization time sums have opposite signs ($\sum d_{J_0} < 0$, $\sum d_{J_1} > 0$).

Woodbury correction: The correction is the sum of per-subject rank-one outer products:

$$W = \sum_{g \in \{\text{trt}, \text{plc}\}} X_i^{(g)'} R^{-1} Z \cdot H \cdot Z' R^{-1} X_i^{(g)}$$

which is generally rank 2 (not rank 1), so $X'\Sigma^{-1}X = X'R^{-1}X - W$.

3.4.2 Numerical implementation

[29] **For general run-in and post-randomization counts the variance has no compact closed form and must be evaluated as a complex ratio.**

- No simple closed-form expression exists for arbitrary J_0 and J_1 .
- The variance depends on the sum-of-squares and sum quantities together with the dimension J .
- The Woodbury correction interacts with the information matrix to produce algebraically complex expressions.

To be completed by rgt.

The variance $\text{Var}(\hat{\gamma})$ for general J_0 and J_1 does not admit a simple closed-form expression. The formula depends on σ^2 and σ_b^2 in a ratio involving S_{J_0} , S_{J_1} , D_{J_0} , D_{J_1} , and J , and the interaction between the Woodbury correction and the information matrix produces algebraically complex expressions.

[30] **A matrix-based R implementation evaluates the exact variance for any configuration, with the closed-form cases as checkpoints.**

- The computation uses the matrix Woodbury identity in R code.
- It returns the exact variance for any (J_0, J_1, J_2) configuration.
- The $J_0 = 0$ and $J_0 = 2$ closed forms serve as verification checkpoints.

To be completed by rgt.

We therefore implement the computation via the matrix-based Woodbury identity in R code, which evaluates the exact variance for any configuration (J_0, J_1, J_2) . The closed-form expressions for $J_0 = 0$ (19) and $J_0 = 2$ (30) serve as verification checkpoints.

3.4.3 Remark: recovery of the Edland and Diggle formulae

[31] **Setting the run-in count to zero collapses the information matrix to a two-parameter block and yields the slope-comparison variance.**

- With $J_0 = 0$ the δ column vanishes, leaving a 2×2 block on (β, γ) .
- Sherman-Morrison cancellations collapse the Woodbury correction against the remaining structure.
- The result is the per-arm variance of the treatment-effect estimator for n participants.

To be completed by rgt.

At $J_0 = 0$, the δ column of X is empty and the information matrix reduces to a 2×2 block on (β, γ) . Carrying the Woodbury algebra through, the Sherman-Morrison cancellations collapse the Woodbury correction against the $(2, 1; 1, 1)$ structure of $X'R^{-1}X$, and the variance of the treatment effect estimator for n participants per arm becomes

$$\text{Var}(\hat{\gamma})|_{J_0=0, J_2=0} = \frac{2}{n} \left[\sigma_b^2 + \frac{\sigma^2}{S_{xx}} \right], \quad S_{xx} = \sum_{j=0}^{J_1} (t_j - \bar{t})^2, \quad (34)$$

[32] **The boundary case recovers the Edland, Diggle, and Frost formulae, confirming that the general result embeds the long-power routines as strict special cases.**

- The expression is the Ard-Edland slope-comparison formula implemented in the longpower edland routine.
- Setting the slope variance to zero gives compound symmetry and recovers the Diggle formula.
- The Frost formula is the two-visit specialization, so the general result quantifies gains accruing only from run-in or common-close observations.

To be completed by rgt.

where the baseline visit at $t_0 = 0$ enters through the change-score transformation. Equation (34) is the slope-comparison formula of⁸ implemented in `longpower::edland.linear.power()`. Setting additionally $\sigma_b^2 = 0$ reduces Σ to compound symmetry and recovers the formula of⁹ used by `longpower::diggle.linear.power()`. The Frost formula (19) is in turn the $J_1 = 2$ specialization of (34), since $S_{xx} = 2t^2$ at that setting. The general- (J_0, J_1, J_2) result derived in the next subsection therefore embeds both `longpower` closed-form routines as strict boundary cases, with the efficiency gains quantified in Section 5 accruing entirely from the information contributed by $J_0 > 0$ or $J_2 > 0$.

3.5 General formula with common close ($J_2 \geq 0$)

[33] **The final extension adds common-close observations after the treatment period to the change-score vector.**

- J_2 common-close observations follow the treatment period.
- All participants are off-treatment during this phase.
- The total dimension becomes $J = J_0 + J_1 + J_2$.

To be completed by rgt.

We now extend to include J_2 common close observations after the treatment period. The total dimension is $J = J_0 + J_1 + J_2$.

During the common close phase (time points $J_1 + 1, \dots, J_1 + J_2$), all participants are off-treatment. The change scores are:

$$c_{i, J_1+s} = [\beta(J_1 + s) + \gamma g_i \cdot J_1 + b_i(J_1 + s)]t + (w_{i, J_1+s} - w_{i0}) \quad s = 1, \dots, J_2 \quad (35)$$

[34] **In the common close the treatment effect persists only as an accumulated offset, fixing the design-matrix entry at the treatment-period value.**

- γ enters only through the effect accumulated during the treatment period.
- It does not contribute to the slope during the common close.
- The γ column therefore equals $J_1 t$ for all common-close observations in the treatment group.

To be completed by rgt.

The treatment effect γ enters the common close change scores only through the accumulated effect during the treatment period ($\gamma g_i \cdot J_1 \cdot t$), not through the slope during the common close. This means the γ column in the design matrix has the value $J_1 t$ for all common close observations in the treatment group.

636 The fixed effects design matrix becomes:

$$X_i^{(\text{trt})} = t \begin{pmatrix} -J_0 & 0 & 0 \\ \vdots & \vdots & \vdots \\ -1 & 0 & 0 \\ 0 & 1 & 1 \\ \vdots & \vdots & \vdots \\ 0 & J_1 & J_1 \\ 0 & J_1 + 1 & J_1 \\ \vdots & \vdots & \vdots \\ 0 & J_1 + J_2 & J_1 \end{pmatrix}$$

637 [35] **The slope and treatment columns of the design matrix**
 638 **encode the post-randomization times and the capped treat-**
 639 **ment exposure respectively.**

- 640 • The β column holds the post-randomization times of the placebo
- 641 slope across all later visits.
- 642 • The γ column rises through the treatment period then stays at
- 643 J_1 during the common close.
- 644 • The γ column is identically zero for placebo participants.

645 To be completed by rgt.

646 *The β column contains the post-randomization time for the placebo slope:*
 647 *$0, \dots, 0, 1, \dots, J_1, J_1 + 1, \dots, J_1 + J_2$. The γ column is $0, \dots, 0, 1, \dots, J_1, J_1, \dots, J_1$*
 648 *for treatment participants and all zeros for placebo.*

649 The random effects design remains $Z_i = t \cdot \mathbf{d}$ where $\mathbf{d} = (-J_0, \dots, -1, 1, \dots, J_1, J_1 +$
 650 $1, \dots, J_1 + J_2)'$.

651 The residual covariance is still $R = a(\mathbf{I}_J + \mathbf{1}_J \mathbf{1}_J')$ with $J = J_0 + J_1 + J_2$.

652 [36] **The common-close extension reuses the same Woodbury**
 653 **computation with only three quantities altered.**

- 654 • The dimension J increases to include the common-close visits.
- 655 • The X matrix changes only in its γ column.
- 656 • The elements of the random-effects vector Z are extended accord-
- 657 ingly.

658 To be completed by rgt.

659 *The Woodbury computation proceeds identically, with the only changes being*
 660 *the dimension J , the structure of the X matrix (specifically the γ column),*
 661 *and the elements of Z .*

662 We implement the general computation in R below.

4 Power and Sample Size Formulas

For a two-sided test at significance level α with power $1 - \beta_{\text{pow}}$, the required number of participants per group is

$$m = \frac{(z_{\alpha/2} + z_{1-\beta_{\text{pow}}})^2 \cdot \text{Var}_1(\hat{\gamma})}{\Delta^2} \quad (36)$$

where $\text{Var}_1(\hat{\gamma})$ denotes the per-participant-pair variance from the formulas derived above, and Δ is the minimum clinically important treatment effect on the rate of change.

Equivalently, the power for a given sample size is

$$1 - \beta_{\text{pow}} = \Phi\left(\frac{\Delta\sqrt{m}}{\sqrt{\text{Var}_1(\hat{\gamma})}} - z_{\alpha/2}\right) \quad (37)$$

The relative efficiency of a design with (J_0, J_1, J_2) observations compared to a standard design with $(0, J_1, 0)$ is

$$\text{RE}(J_0, J_1, J_2) = \frac{\text{Var}_1(\hat{\gamma})|_{0, J_1, 0}}{\text{Var}_1(\hat{\gamma})|_{J_0, J_1, J_2}} \quad (38)$$

Values greater than 1 indicate that the run-in/common close design requires fewer participants.

[37] A single run-in observation can be counterproductive because the extra slope parameter it introduces may offset the information it contributes.

- Adding one run-in observation introduces the parameter δ , enlarging the fixed-effect dimension.
- This added degree of freedom can make the one-run-in variance exceed the no-run-in variance for some parameters.
- Designers should therefore plan either zero or at least two run-in observations, never exactly one.

To be completed by rgt.

A practical caveat: adding a single run-in observation ($J_0 = 1$) introduces the run-in slope parameter δ into the model, increasing the dimension of β from 2 to 3. This additional degree of freedom can offset the information gained from the run-in observation, so that $\text{Var}(\hat{\gamma})|_{J_0=1}$ may exceed $\text{Var}(\hat{\gamma})|_{J_0=0}$ for some parameter configurations. This effect, noted in a different context by¹⁵, implies that trial designers should plan either zero or at least two run-in observations, not one.

5 Numerical Examples

5.1 Verification of closed-form results

[38] The closed-form expressions are validated against the matrix computation in two representative design configurations.

- The verification confirms agreement between the analytic and matrix-based variances.
- The Frost standard design uses the two-parameter model with two post-baseline visits.
- The two-run-in case exercises the three-parameter model.

To be completed by rgt.

We verify that the closed-form expressions agree with the matrix computation. The Frost et al. (2008) result corresponds to the standard design with $J_0 = 0$, $J_1 = 2$ (two post-baseline visits at times t and $2t$), which under the two-parameter model yields $\text{Var}(\hat{\gamma}) = (\sigma^2 + 2\sigma_b^2 t^2)/t^2$. The $J_0 = 2$, $J_1 = 1$ case uses the three-parameter model.

```
a <- 1.0
b <- 0.5
tt <- 1.0

v_frost_matrix <- var_gamma_matrix(0, 2, 0, a, b, tt)
v_frost_closed <- var_gamma_frost(a, b, tt)

v_r2_matrix <- var_gamma_matrix(2, 1, 0, a, b, tt)
v_r2_closed <- var_gamma_r2(a, b, tt)

cat('Frost (r=0, k=2) case:\n')
```

```
## Frost (r=0, k=2) case:
```

```
cat('  Matrix:', round(v_frost_matrix, 6), '\n')
```

```
## Matrix: 2
```

```
cat('  Closed:', round(v_frost_closed, 6), '\n')
```

```
## Closed: 2
```

```
cat('  Match:',
    all.equal(v_frost_matrix, v_frost_closed), '\n\n')
```

```
## Match: TRUE
```

```
cat('r=2, k=1 case:\n')
```

```
## r=2, k=1 case:
```

Table 1: Required sample size per group ($J_1 = 2$, $J_2 = 0$, $\Delta = 0.5$, 90% power, two-sided $\alpha = 0.05$) as a function of the number of run-in observations J_0 , for three levels of between-subject slope variability.

J_0	Low	Medium	High
0	51	127	463
1	44	119	214
2	44	91	113
3	43	70	76
4	42	56	59
5	40	48	49
6	38	42	43

```

cat('  Matrix:', round(v_r2_matrix, 6), '\n')
711 ##    Matrix: 4.666667
cat('  Closed:', round(v_r2_closed, 6), '\n')
712 ##    Closed: 4.666667
cat('  Match:',
    all.equal(v_r2_matrix, v_r2_closed), '\n\n')
713 ##    Match: TRUE
cat('Variance comparison across designs:\n')
714 ## Variance comparison across designs:
for (rr in 0:4) {
  v <- var_gamma_matrix(rr, 2, 0, a, b, tt)
  cat(sprintf('  r=%d, k=2: Var = %.4f\n', rr, v))
}
715 ##    r=0, k=2: Var = 2.0000
716 ##    r=1, k=2: Var = 2.0000
717 ##    r=2, k=2: Var = 1.7910
718 ##    r=3, k=2: Var = 1.5000
719 ##    r=4, k=2: Var = 1.2651

```

720 5.2 Sample size as a function of run-in observations

721 5.3 Sample size as a function of common close observations

722 5.4 Relative efficiency surface

723 5.5 Application: Alzheimer's disease imaging trial

724 [39] The Alzheimer's application uses MIRIAD-derived pa-

Table 2: Required sample size per group ($J_0 = 2$, $J_1 = 2$, $\Delta = 0.5$, 90% power) as a function of common close observations J_2 .

J_2	Low	Medium	High
0	44	91	113
1	34	69	80
2	29	48	52
3	25	35	37
4	22	27	28

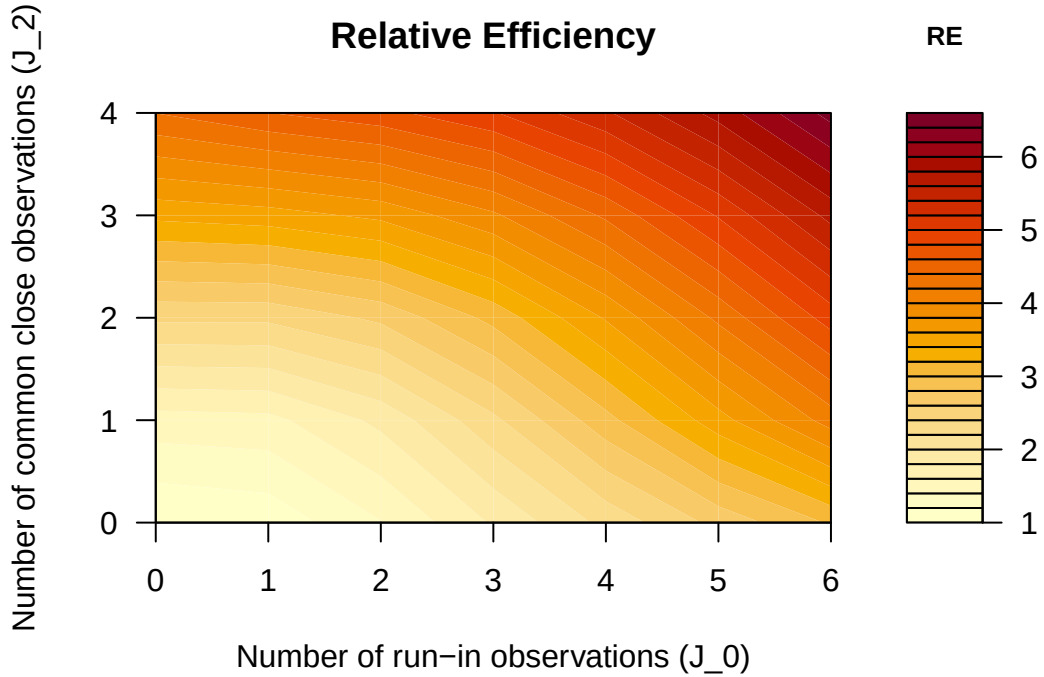


Figure 1: Relative efficiency of run-in and common close designs compared to the standard design ($J_0 = 0$, $J_2 = 0$) with the same number of post-randomization observations ($J_1 = 2$). Parameters: $\sigma^2 = 1$, $\sigma_b^2 = 1$, $t = 1$.

Table 3: Sample size per group for AD imaging trial ($J_1 = 2$ post-randomization visits, $\Delta = 0.25$, 90% power). Top 15 configurations by sample size.

J_0	J_1	J_2	m per group	Total visits
4	2	2	49	9
3	2	2	58	8
4	2	1	60	8
2	2	2	69	7
3	2	1	77	7
0	2	2	77	5
1	2	2	77	6
4	2	0	78	7
3	2	0	100	6
2	2	1	104	6
1	2	1	138	5
2	2	0	145	5
0	2	1	150	4
1	2	0	247	4

rameters with the correctly combined within-visit residual,
which is essential to reproduce Frost’s reported sample sizes.

- The slope variance and combined within-visit residual are taken from Frost’s MIRIAD estimates.
- The example assumes a one-year observation interval and a treatment effect of one quarter.
- Using only the small measurement-error term understates the residual by two orders of magnitude and yields degenerate sample sizes.

To be completed by rgt.

Using variance parameters estimated from the MIRIAD dataset²: $\sigma_b^2 = 0.9745$ (between-subject slope variance) and a combined within-visit residual $\sigma^2 = \sigma_u^2 + \sigma_\varepsilon^2 = 0.3338 + 0.0025 = 0.3363$, formed per Eq. (2) from the within-subject between-visit component ($\sigma_u^2 = 0.3338$) and the additional between-scan measurement error ($\sigma_\varepsilon^2 = 0.0025$) of Frost’s Table II. We take observation interval $t = 1$ year and treatment effect $\Delta = 0.25$ (25% reduction in atrophy rate relative to placebo rate of approximately 1.0% brain volume loss per year). Using the combined residual, the conventional design reproduces Frost’s reported sample sizes (Table III); using only the 0.0025 measurement-error term, as in earlier drafts, understates the residual by two orders of magnitude and yields degenerate sample sizes.

AD imaging trial sample sizes (per group):

Standard (r=0, k=2, f=0): 385

Run-in (r=2, k=2, f=0): 145 (62% reduction)

749 ## Full (r=2, k=2, f=2): 69 (82% reduction)

750 6 Discussion

751 [40] The work formalizes and extends Frost’s framework with
752 explicit variance formulas for run-in and common-close de-
753 signs.

- 754 • The results provide explicit treatment-effect variance formulas.
- 755 • They cover designs incorporating both run-in and common-close
- 756 observations.
- 757 • Three substantive points warrant further discussion.

758 To be completed by rgt.

759 *The results presented here formalize and extend the framework of² by provid-*
760 *ing explicit variance formulas for the treatment effect estimator under designs*
761 *that incorporate run-in and common close observations. Three points merit*
762 *discussion.*

763 [41] The benefit of run-in observations is governed by the
764 slope-to-noise ratio and is largest for outcomes such as neu-
765 roimaging.

- 766 • The efficiency gain depends on the ratio of slope variance to resid-
- 767 ual variance.
- 768 • When that ratio is large, even one or two run-in observations cut
- 769 sample sizes substantially.
- 770 • For cognitive outcomes with a less favorable ratio, gains are mod-
- 771 est and must be weighed against longer study duration.

772 To be completed by rgt.

773 *When run-in observations are most beneficial. The efficiency gain*
774 *from run-in observations depends on the ratio $\sigma_b^2 t^2 / \sigma^2$. When between-subject*
775 *slope variability is large relative to residual variance (as in neuroimaging*
776 *outcomes), even one or two run-in observations yield substantial sample size*
777 *reductions. For cognitive outcomes, where this ratio is typically less favorable,*
778 *the gains are more modest and the cost of extended study duration must be*
779 *weighed against the sample size reduction.*

780 [42] Common-close observations sharpen the treatment-effect
781 estimator at low cost, with diminishing returns beyond the
782 first few visits.

- 783 • They inform individual trajectories without keeping participants
- 784 on treatment.
- 785 • Both groups contribute to slope-variability estimation once treat-
- 786 ment has ceased.
- 787 • Marginal benefit diminishes, but the first one or two visits give
- 788 meaningful reductions when the slope-to-noise ratio is large.

789 To be completed by rgt.

790 *Common close as a low-cost efficiency gain.* The common close phase
791 provides information about individual trajectories without requiring partic-
792 ipants to remain on treatment. Since treatment has ceased, both groups
793 contribute to estimation of the underlying slope variability, effectively tight-
794 ening the precision of the treatment effect estimator. The marginal benefit
795 of each additional common close observation diminishes (as with run-in ob-
796 servations), but the first one or two common close visits typically provide
797 meaningful sample size reductions, particularly when $\sigma_b^2 t^2 / \sigma^2$ is large.

798 [43] **The derived formulas are consistent with Wang’s two-**
799 **period model while making the three phases explicit and**
800 **separately optimizable.**

- 801 • Wang’s common-slope scenario corresponds to setting the run-in
802 and placebo slopes equal.
- 803 • Their different-slope scenario corresponds to the general case
804 treated here.
- 805 • The present formulation separates run-in, treatment, and
806 common-close phases, enabling per-phase optimization.

807 To be completed by rgt.

808 *Relationship to the two-period model.* The formulas derived here are
809 consistent with the two-period linear mixed effects model of³. Their Scenario
810 1 (common slope across periods) corresponds to setting $\delta = \beta$ in our model,
811 while their Scenario 2 (different slopes) corresponds to the general case. The
812 advantage of our formulation is the explicit separation of run-in, treatment,
813 and common close phases, enabling direct optimization of the number of
814 observations in each phase.

815 [44] **The derivations rest on simplifying assumptions whose**
816 **relaxation is feasible but typically forfeits closed-form sim-**
817 **plicity.**

- 818 • They assume equally spaced observations, linear trajectories, and
819 a random intercept-plus-slope structure.
- 820 • Extensions to unequal spacing, nonlinear progression, or richer
821 random effects remain within the Woodbury framework but lose
822 simple closed forms.
- 823 • No dropout is assumed, though the Dawson-Lagakos pattern-
824 mixture approach can adjust for anticipated attrition.

825 To be completed by rgt.

826 *Limitations.* The derivations assume equally spaced observations, linear
827 trajectories, and a random intercept plus random slope covariance structure.
828 Extensions to unequal spacing, nonlinear progression, and more complex ran-
829 dom effects structures (e.g., random intercept-slope correlation) are possible
830 within the same Woodbury identity framework but yield more complex expres-
831 sions that may not admit simple closed forms. The formulas also assume no

832 *dropout; in practice, the Dawson-Lagakos pattern-mixture approach² can be*
833 *applied to adjust for anticipated attrition.*

834 7 Conflict of interest

835 The author declares no conflicts of interest.

836 8 Funding

837 This work received no specific funding.

838 9 Data availability statement

839 [45] **All artefacts supporting the paper are openly available in**
840 **the project repository.**

- 841 • These include the R package, the four companion manuscripts,
842 and the MATLAB derivation scripts.
- 843 • The shared bibliography is likewise available.
- 844 • Specific paths are enumerated in the Reproducibility section.

845 *To be completed by rgt.*

846 *The `runinpower` R package, the four companion manuscripts of this com-*
847 *pendium, the MATLAB symbolic-derivation scripts, and the shared bibliogra-*
848 *phy are openly available in the project repository. Specific paths are listed in*
849 *the Reproducibility section below.*

850 10 Reproducibility

851 [46] **The manuscript build relies on the in-package source to-**
852 **gether with a small set of testing and rendering packages.**

- 853 • Rendering used R version 4.4.x.
- 854 • The principal dependencies are the in-package source plus
855 `tinytest`, `rmarkdown`, and `bookdown`.
- 856 • Supporting MATLAB symbolic derivations reside in the analysis
857 scripts directory.

858 *To be completed by rgt.*

859 *This manuscript was rendered with R 4.4.x. Principal packages used*
860 *are the in-package `runinpower` source (nine source files: `allocation`,*
861 *`ancova`, `averaged`, `covariance`, `covariance-ar1`, `design`, `power`, `replicated`,*
862 *`variance`), `tinytest` (testing harness), and `rmarkdown` plus `bookdown`*
863 *for the manuscript build. MATLAB symbolic derivations supporting the*
864 *Woodbury-based variance formulas are at `analysis/scripts/`.*

865 [47] **The variance results are deterministic, and any Monte**
866 **Carlo benchmarking uses a fixed seed for reproducibility.**

- 867 • Given fixed input parameters the formulas exercise no random-
868 number generation.
- 869 • Monte Carlo benchmarking against the analytic formulas is the
870 only stochastic component.
- 871 • That benchmarking driver fixes the seed at its entry point.

872 To be completed by rgt.

873 *Random-number generator.* The variance formulas are deterministic
874 given fixed input parameters; no RNG is exercised. Where Monte Carlo
875 benchmarking against the analytic formulas is reported, the driver sets
876 `set.seed(20260418)` at its entry point.

877 [48] **The repository organizes the package source, derivation**
878 **scripts, companion manuscripts, and bibliography into well-**
879 **defined locations.**

- 880 • The R package source and the MATLAB scripts each have a
881 dedicated directory.
- 882 • The three companion manuscripts share a report subtree.
- 883 • This manuscript source and the shared bibliography sit in the
884 run-in power report directory.

885 To be completed by rgt.

886 *Data and code availability.* The R package source is at `R/`. MAT-
887 LAB scripts are at `analysis/scripts/`. The companion manuscripts are at
888 `analysis/report/{02,03,04}-*/`. The manuscript source and shared bibli-
889 ography (`./references.bib`) are at `analysis/report/01-runin-power/`.

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